

---

---

**Energy management systems —  
Requirements for bodies providing  
audit and certification of energy  
management systems**

*Systèmes de management de l'énergie — Exigences pour les  
organismes procédant à l'audit et à la certification de systèmes de  
management de l'énergie*





**COPYRIGHT PROTECTED DOCUMENT**

© ISO 2021

All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office  
CP 401 • Ch. de Blandonnet 8  
CH-1214 Vernier, Geneva  
Phone: +41 22 749 01 11  
Email: [copyright@iso.org](mailto:copyright@iso.org)  
Website: [www.iso.org](http://www.iso.org)

Published in Switzerland

# Contents

Page

|                                                                                             |            |
|---------------------------------------------------------------------------------------------|------------|
| <b>Foreword</b>                                                                             | <b>v</b>   |
| <b>Introduction</b>                                                                         | <b>vii</b> |
| <b>1 Scope</b>                                                                              | <b>1</b>   |
| <b>2 Normative references</b>                                                               | <b>1</b>   |
| <b>3 Terms and definitions</b>                                                              | <b>1</b>   |
| <b>4 Principles</b>                                                                         | <b>3</b>   |
| <b>5 General requirements</b>                                                               | <b>3</b>   |
| <b>6 Structural requirements</b>                                                            | <b>3</b>   |
| <b>7 Resource requirements</b>                                                              | <b>3</b>   |
| 7.1 Technical competence                                                                    | 3          |
| 7.2 Personnel involved in the certification activities                                      | 3          |
| 7.2.1 General                                                                               | 3          |
| 7.2.2 Energy specific terminology                                                           | 4          |
| 7.2.3 Energy principles                                                                     | 5          |
| 7.2.4 Energy related legal requirements                                                     | 5          |
| 7.2.5 Knowledge of ISO 50001 requirements                                                   | 5          |
| 7.2.6 Energy performance indicators, energy baseline, relevant variables and static factors | 6          |
| 7.2.7 Common energy using systems                                                           | 6          |
| 7.2.8 Energy performance improvement                                                        | 7          |
| 7.2.9 Principles of data collection, monitoring, measurement, analysis and evaluation       | 7          |
| 7.3 Use of individual external auditors and external technical experts                      | 7          |
| 7.4 Personnel records                                                                       | 7          |
| 7.5 Outsourcing                                                                             | 8          |
| <b>8 Information requirements</b>                                                           | <b>8</b>   |
| 8.1 Public information                                                                      | 8          |
| 8.2 Certification documents                                                                 | 8          |
| 8.3 Reference to certification and use of marks                                             | 8          |
| 8.4 Confidentiality                                                                         | 8          |
| 8.5 Information exchange between a certification body and its clients                       | 8          |
| <b>9 Process requirements</b>                                                               | <b>8</b>   |
| 9.1 Pre-certification activities                                                            | 8          |
| 9.1.1 Application                                                                           | 8          |
| 9.1.2 Application review                                                                    | 8          |
| 9.1.3 Audit programme                                                                       | 8          |
| 9.1.4 Determining audit time                                                                | 9          |
| 9.1.5 Multi-site sampling                                                                   | 10         |
| 9.1.6 Multiple management system standards                                                  | 10         |
| 9.2 Planning audits                                                                         | 10         |
| 9.2.1 Determining audit objectives, scope and criteria                                      | 10         |
| 9.2.2 Audit team selection and assignments                                                  | 10         |
| 9.2.3 Audit plan                                                                            | 10         |
| 9.3 Initial certification                                                                   | 10         |
| 9.3.1 Stage 1                                                                               | 10         |
| 9.3.2 Stage 2                                                                               | 11         |
| 9.4 Conducting audits                                                                       | 11         |
| 9.4.1 General                                                                               | 11         |
| 9.4.2 Conducting the opening meeting                                                        | 11         |
| 9.4.3 Communication during the audit                                                        | 11         |
| 9.4.4 Obtaining and verifying information                                                   | 11         |
| 9.4.5 Identifying and recording audit findings                                              | 11         |

|                     |                                                                      |           |
|---------------------|----------------------------------------------------------------------|-----------|
| 9.4.6               | Preparing audit conclusions.....                                     | 11        |
| 9.4.7               | Conducting the closing meeting.....                                  | 11        |
| 9.4.8               | Audit report.....                                                    | 11        |
| 9.5                 | Certification decision.....                                          | 12        |
| 9.5.1               | General.....                                                         | 12        |
| 9.5.2               | Actions prior to making a decision.....                              | 12        |
| 9.5.3               | Information for granting initial certification.....                  | 12        |
| 9.5.4               | Information for granting recertification.....                        | 12        |
| 9.6                 | Maintaining certification.....                                       | 12        |
| 9.6.1               | General.....                                                         | 12        |
| 9.6.2               | Surveillance audit.....                                              | 13        |
| 9.6.3               | Recertification.....                                                 | 13        |
| 9.6.4               | Special audits.....                                                  | 13        |
| 9.6.5               | Suspending, withdrawing or reducing the scope of certification.....  | 13        |
| 9.7                 | Appeals.....                                                         | 13        |
| 9.8                 | Complaints.....                                                      | 13        |
| 9.9                 | Client records.....                                                  | 13        |
| <b>10</b>           | <b>Management system requirements for certification bodies .....</b> | <b>13</b> |
| <b>Annex A</b>      | <b>(normative) EnMS audit time .....</b>                             | <b>14</b> |
| <b>Annex B</b>      | <b>(normative) Multi-site organizations .....</b>                    | <b>19</b> |
| <b>Annex C</b>      | <b>(informative) Energy performance improvement .....</b>            | <b>27</b> |
| <b>Annex D</b>      | <b>(informative) Examples of audit calculations .....</b>            | <b>28</b> |
| <b>Bibliography</b> | <b>.....</b>                                                         | <b>33</b> |

## Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see [www.iso.org/directives](http://www.iso.org/directives)).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see [www.iso.org/patents](http://www.iso.org/patents)).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see [www.iso.org/iso/foreword.html](http://www.iso.org/iso/foreword.html).

This document was prepared by Technical Committee ISO/TC 301, *Energy management and energy savings*, in collaboration with the ISO Committee on conformity assessment (CASCO).

This second edition cancels and replaces the first edition (ISO 50003:2014), which has been technically revised. The main changes compared with the previous edition are as follows:

- the definitions have been updated to include the audit time, the duration of the audit and terms related to multi-site audits;
- the phrase “maintained documented information” has been used to represent procedures, work instructions or other forms of documents that provide the who, what, when, how or why information;
- the phrase “retained documented information” or “record of audit evidence” has been used to represent records that demonstrate or provide evidence of the execution of a requirement;
- the structure has been updated to align with ISO/IEC 17021-1:2015;
- the phrase “man-days” has been changed to “audit days”;
- for audit day calculations, the number of energy types have been changed to those that comprise at least 80 % of total consumption;
- the weighted values for complexity have been modified;
- the sampling requirements for multi-site EnMS have been updated;
- the use of IAF MD documents as they relate to [Annexes A](#) and [B](#) has been clarified;
- the information on EnMS effective personnel has been clarified in [A.2](#);
- [Tables A.3](#) and [A.4](#) have been modified to refer to audit time rather than the duration of the audit;
- the technical areas have been removed and requirements for technical competency added.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at [www.iso.org/members.html](http://www.iso.org/members.html).

## Introduction

This document is intended to be used in conjunction with ISO/IEC 17021-1:2015.

In addition to the requirements of ISO/IEC 17021-1:2015, this document specifies requirements for the specific technical area of energy management systems (EnMS) that are needed to ensure the effectiveness of the audit and certification, while supporting the efforts of the organization to continually improve energy performance including energy efficiency, energy use and energy consumption, and the EnMS. In particular, this document addresses the additional requirements necessary for the audit process. It covers the planning process, the initial certification audit, conducting the on-site audit, auditor competence, audit time and multi-site sampling. The structure of this document follows that of ISO/IEC 17021-1:2015. [Annexes A](#) and [B](#) are normative while [Annexes C](#) and [D](#) provide additional information to complement ISO/IEC 17021-1:2015.

This document deals with EnMS audits for certification purposes, but it does not deal with energy audits where the purpose is to establish a systematic analysis of energy consumption and energy use, and which are defined in ISO 50002.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

In this document, references to the word “site” can be taken as either singular, meaning one permanent site (physical or virtual) or temporary site (physical or virtual), or can be plural, meaning more than one permanent site or temporary site, unless otherwise specified.



# Energy management systems — Requirements for bodies providing audit and certification of energy management systems

## 1 Scope

This document specifies requirements for competence, consistency and impartiality in the auditing and certification of ISO 50001 energy management systems (EnMS) for bodies providing these services. In order to ensure the effectiveness of EnMS auditing, this document addresses the auditing process, the competence requirements for the personnel involved in the certification process for EnMS, the audit time and multi-site sampling.

## 2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO/IEC 17021-1:2015, *Conformity assessment — Requirements for bodies providing audit and certification of management systems — Part 1: Requirements*

ISO 50001, *Energy management systems — Requirements with guidance for use*

## 3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO/IEC 17021-1:2015, ISO 50001 and the following apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- ISO Online browsing platform: available at <https://www.iso.org/obp>
- IEC Electropedia: available at <http://www.electropedia.org/>

### 3.1

#### **audit evidence**

documented information, statements of fact or other information, which are relevant to the audit criteria and verifiable

Note 1 to entry: Audit evidence can be qualitative or quantitative.

[SOURCE: ISO 9000:2015, 3.13.8, modified — The word “records” has been replaced by “documented information” and Note 1 to entry has been added.]

### 3.2

#### **duration of the audit**

part of *audit time* (3.3) spent conducting audit activities from the opening meeting to the closing meeting, inclusive

Note 1 to entry: Audit activities normally include:

- conducting the opening meeting;
- performing document review while conducting the audit;

- communicating during the audit;
- assigning roles and responsibilities of guides and observers;
- collecting and verifying information;
- generating audit findings;
- preparing audit conclusions;
- conducting the closing meeting.

[SOURCE: ISO/IEC 17021-1:2015, 3.17, modified — The term “duration of management system certification audits” has been replaced by “duration of the audit”.]

### 3.3 audit time

time needed to plan and accomplish a complete and effective audit of the client organization's management system

[SOURCE: ISO/IEC 17021-1:2015, 3.16]

### 3.4 central function

function that is responsible for and controls the EnMS of a *multi-site organization* (3.7)

Note 1 to entry: The central function is not necessarily operating from the headquarters or a single site.

Note 2 to entry: The authority of the central function is derived from top management. The central function has authority over every site regarding the EnMS.

### 3.5 EnMS effective personnel

personnel who materially contribute to the effectiveness of the EnMS or impact energy performance

Note 1 to entry: EnMS effective personnel is not necessarily the total number of employees (headcount).

Note 2 to entry: The EnMS effective personnel number is a factor used to determine the *audit time* (3.3).

### 3.6 major nonconformity

<energy management system> nonconformity that affects the capability of the EnMS to achieve the intended results

Note 1 to entry: Classifying nonconformities as major could be as follows:

- *audit evidence* (3.1) that energy performance improvement was not achieved;
- a significant doubt that effective process control is in place;
- a number of minor nonconformities associated with the same requirements or issue could demonstrate a systemic failure and thus constitute a major nonconformity.

[SOURCE: ISO/IEC 17021-1:2015, 3.12, modified — The words “management system” have been replaced by “EnMS” and the first bullet has been added to Note 1 to entry.]

### 3.7 multi-site organization

organization having an identified *central function* (3.4) and more than one site (permanent or temporary) with a common management system

Note 1 to entry: The EnMS of a multi-site organization is established, implemented, maintained and subject to internal audits planned by the central function.

**3.8****permanent site**

location (physical or virtual) where a client organization performs work or provides a service on a continual basis

[SOURCE: ISO/IEC TS 17023:2013, 3.4, modified — The word “continuing” has been replaced by “continual”.]

**3.9****temporary site**

location (physical or virtual) where a client organization performs specific work or provides a service for a finite period of time and which is not intended to become a *permanent site* ([3.8](#))

EXAMPLE Construction site; road site.

[SOURCE: ISO/IEC TS 17023:2013, 3.5, modified — Example has been added.]

**4 Principles**

Principles are not requirements. The principles of ISO/IEC 17021-1:2015, Clause 4, apply.

**5 General requirements**

All the requirements defined in ISO/IEC 17021-1:2015 and in this document shall be applied to the EnMS auditing process.

This document defines the requirements for the determination of audit time for an EnMS in [Annex A](#).

This document defines the requirements for the multi-site sampling of a client organization’s EnMS in [Annex B](#).

**6 Structural requirements**

The requirements of ISO/IEC 17021-1:2015, Clause 6, apply. In addition, when safeguarding impartiality, the certification body shall manage, as a potential conflict of interest, energy audits and/or other energy related services conducted by the auditor or the certification body within the last two years.

**7 Resource requirements****7.1 Technical competence**

The requirements of ISO/IEC 17021-1:2015, Clause 7, apply. Basic knowledge of the client’s business and typical business processes knowledge is defined in ISO/IEC 17021-1:2015. (See ISO/IEC 17021-1:2015, Table A.1.) In addition, the technical competence requirements for the audit team and personnel involved in the EnMS certification process are defined in [7.2](#).

NOTE This document does not apply technical sectors when defining technical competence.

**7.2 Personnel involved in the certification activities****7.2.1 General**

The competencies shall include a level of generic competence described in ISO/IEC 17021-1:2015 as well as the EnMS technical knowledge described in [Table 1](#), where “X” indicates that the certification body shall define the criteria and depth of knowledge. The certification body personnel shall have the competencies defined in [Table 1](#).

In addition to the technical knowledge requirements specified in [Table 1](#), the certification body shall define criteria, including the knowledge and skills of the audit team that is necessary for the client.

**Table 1 — Required EnMS technical knowledge**

| Knowledge                                                                                           | Certification functions                                                                                                                             |                                                            |                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------|
|                                                                                                     | Conducting the application review to determine the required audit team competence, to select the audit team members and to determine the audit time | Reviewing audit reports and making certification decisions | Auditing                      |
| Energy specific terminology                                                                         | X ( <a href="#">7.2.2.3</a> )                                                                                                                       | X ( <a href="#">7.2.2.2</a> )                              | X ( <a href="#">7.2.2.1</a> ) |
| Energy principles                                                                                   | X ( <a href="#">7.2.3.3</a> )                                                                                                                       | X ( <a href="#">7.2.3.2</a> )                              | X ( <a href="#">7.2.3.1</a> ) |
| Energy-related legal requirements                                                                   | —                                                                                                                                                   | X ( <a href="#">7.2.4.2</a> )                              | X ( <a href="#">7.2.4.1</a> ) |
| Knowledge of ISO 50001 requirements                                                                 | X ( <a href="#">7.2.5.3</a> )                                                                                                                       | X ( <a href="#">7.2.5.2</a> )                              | X ( <a href="#">7.2.5.1</a> ) |
| Energy performance indicators (EnPIs), energy baseline (EnB), relevant variables and static factors | —                                                                                                                                                   | X ( <a href="#">7.2.6.2</a> )                              | X ( <a href="#">7.2.6.1</a> ) |
| Common energy using systems                                                                         | —                                                                                                                                                   | X ( <a href="#">7.2.7.2</a> )                              | X ( <a href="#">7.2.7.1</a> ) |
| Energy performance improvement                                                                      | —                                                                                                                                                   | X ( <a href="#">7.2.8.2</a> )                              | X ( <a href="#">7.2.8.1</a> ) |
| Principles of data collection and of monitoring, measuring and evaluating data                      | —                                                                                                                                                   | X ( <a href="#">7.2.9.2</a> )                              | X ( <a href="#">7.2.9.1</a> ) |

## 7.2.2 Energy specific terminology

### 7.2.2.1 Auditing

The certification body auditors shall have knowledge of the terminology from ISO 50001.

Certification bodies may include terminology from ISO 50002, ISO 50006, ISO 50015 or ISO 50047 for their auditors, as determined appropriate.

### 7.2.2.2 Reviewing audit reports and making certification decisions

The personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of the terminology from ISO 50001.

Certification bodies may include additional terminology as determined appropriate for those reviewing audit reports and making certification decisions.

### 7.2.2.3 Conducting the application review to determine the required audit team competence, to select the audit team members and to determine the audit time

The personnel responsible for conducting the application review, selecting the audit team, determining the needed audit competence and determining the audit time shall have knowledge of the terms and definitions from ISO 50001.

### **7.2.3 Energy principles**

#### **7.2.3.1 Auditing**

The certification body auditor shall have knowledge of energy principles. Energy principles shall include at a minimum: types of energy, energy uses, energy conversion, calculation of energy in different units (e.g. kWh to TJ) and power.

The certification body audit team shall have knowledge of the following principles:

- fuel combustion;
- energy flow;
- energy losses;
- energy efficiency;
- energy balance.

#### **7.2.3.2 Reviewing audit reports and making certification decisions**

The personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of the energy principles, including types of energy, energy uses and energy conversion.

#### **7.2.3.3 Conducting the application review to determine the required audit team competence, to select the audit team members and to determine the audit time**

The personnel responsible for conducting the application review, selecting the audit team, determining the needed audit competence and determining the audit time shall have knowledge of the energy principles, including types of energy, energy-uses, energy conversion and calculation of energy in different units (e.g. kWh to TJ).

### **7.2.4 Energy related legal requirements**

#### **7.2.4.1 Auditing**

The certification body audit team shall have knowledge of the basic legal structure related to energy and energy consumption.

#### **7.2.4.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of the basic legal structure related to energy and energy consumption.

### **7.2.5 Knowledge of ISO 50001 requirements**

#### **7.2.5.1 Auditing**

The certification body auditor shall have knowledge of ISO 50001 requirements.

#### **7.2.5.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of ISO 50001 requirements.

**7.2.5.3 Conducting the application review to determine the required audit team competence, to select the audit team members and to determine the audit time**

The team of personnel responsible for conducting the application review, selecting the audit team, determining the needed audit competence and determining the audit time shall have knowledge of ISO 50001 requirements.

**7.2.6 Energy performance indicators, energy baseline, relevant variables and static factors**

**7.2.6.1 Auditing**

The certification body auditor shall have knowledge of EnPIs, EnPI value, EnB, relevant variables, static factors, and their use in an EnMS and in demonstrating energy performance improvement including normalization techniques.

The certification body audit team shall have knowledge of the use of models such as ratios and simple linear regression to carry out normalization of EnPI value(s) and their corresponding EnB(s).

NOTE See ISO 50006 and ISO 50047 for more details.

**7.2.6.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of EnPIs, EnPI value, EnB, relevant variables, static factors and their use in demonstrating energy performance improvement in an EnMS.

**7.2.7 Common energy using systems**

**7.2.7.1 Auditing**

The certification body audit team shall have knowledge of energy using systems, for example:

- boiler and thermal fluid systems;
- refrigeration, heating, ventilation and air conditioning;
- mechanical systems (e.g. motors, fans, pumps);
- thermal, not building envelope (e.g. furnace);
- compressed air;
- lighting;
- additional systems may be included as determined appropriate by the certification body.

**7.2.7.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of energy using systems, for example:

- boiler and thermal fluid systems;
- refrigeration, heating, ventilation and air conditioning;
- mechanical systems (e.g. motors, fans, pumps);
- thermal, not building envelope (e.g. furnace);
- compressed air;

- lighting;
- additional systems may be included as determined appropriate by the certification body.

## **7.2.8 Energy performance improvement**

### **7.2.8.1 Auditing**

The certification body auditor shall have knowledge of potential energy performance improvements in energy using systems.

The certification body audit team shall have knowledge of the application of current technology used to achieve energy performance improvement.

### **7.2.8.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of potential energy performance improvements in energy using systems and the application of current technology used to achieve energy performance improvements.

## **7.2.9 Principles of data collection, monitoring, measurement, analysis and evaluation**

### **7.2.9.1 Auditing**

The certification body auditor shall have knowledge of the types of data that are typically included in the EnMS.

The certification body auditor shall have knowledge of the typical monitoring, measurement and evaluation used in an EnMS.

The certification body auditor shall have knowledge of the use of evaluation of improvement in energy performance by comparing EnPI(s) values against their corresponding normalized EnB(s).

NOTE See ISO 50006 and ISO 50015 for additional details.

### **7.2.9.2 Reviewing audit reports and making certification decisions**

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of the types of data that are typically included in the EnMS.

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of the typical monitoring, measurement and evaluation used in an EnMS.

The team of personnel responsible for reviewing audit reports and making certification decisions within the certification body shall have knowledge of evaluation of improvement in energy performance by comparing EnPI(s) values against their corresponding normalized EnB(s).

## **7.3 Use of individual external auditors and external technical experts**

The requirements of ISO/IEC 17021-1:2015, 7.3, apply.

## **7.4 Personnel records**

The requirements of ISO/IEC 17021-1:2015, 7.4, apply.

## **7.5 Outsourcing**

The requirements of ISO/IEC 17021-1:2015, 7.5, apply.

## **8 Information requirements**

### **8.1 Public information**

The requirements of ISO/IEC 17021-1:2015, 8.1, apply.

### **8.2 Certification documents**

The requirements of ISO/IEC 17021-1:2015, 8.2, apply.

The certification document(s) shall identify the scope and boundary(ies) of the EnMS, which may include activities, facilities and processes related to the EnMS.

The scope and boundary(ies) can include an entire entity with multi-sites, a site, or a subset or subsets within a site such as a building, a facility or a process.

The scope of the certification statement shall not be misleading or include any claims (e.g. 3.5 % improvement in electricity consumption).

NOTE ISO/IEC 17029 contains the requirements for verification and/or validation bodies.

### **8.3 Reference to certification and use of marks**

The requirements of ISO/IEC 17021-1:2015, 8.3, apply.

### **8.4 Confidentiality**

The requirements of ISO/IEC 17021-1:2015, 8.4, apply.

### **8.5 Information exchange between a certification body and its clients**

The requirements of ISO/IEC 17021-1:2015, 8.5, apply.

## **9 Process requirements**

### **9.1 Pre-certification activities**

#### **9.1.1 Application**

- a) The requirements of ISO/IEC 17021-1:2015, 9.1.1, apply.
- b) The client application shall include the relevant details to ensure that the necessary information for calculating the audit time, based on [Annex A](#), is available.
- c) Application requirements for multi-site audits are addressed in [B.5.2](#).

#### **9.1.2 Application review**

The requirements of ISO/IEC 17021-1:2015, 9.1.2, apply.

#### **9.1.3 Audit programme**

The requirements of ISO/IEC 17021-1:2015, 9.1.3, apply.

## 9.1.4 Determining audit time

### 9.1.4.1 General

The requirements of ISO/IEC 17021-1:2015, 9.1.4, apply.

In addition, in determining the audit time, the certification body shall include the following factors:

- a) the number of EnMS effective personnel (see [A.2](#));
- b) the number of energy types (see [A.3](#));
- c) the annual energy consumption (TJ);
- d) the number of significant energy uses (SEUs).

The audit time shall be determined by using [Table A.3](#) for the initial audit and [Table A.4](#) for surveillance and recertification. The calculation method is described in [Annex A](#).

Audit time includes the time spent on-site at a client's site (permanent or temporary) and time spent off-site carrying out planning, document review, interacting with client personnel and report writing.

Travel (en route or between sites) and any breaks shall not be included in the on-site duration of the audits.

The justification and calculations for the determination of the audit time, including duration of the audit shall be recorded, retained as documented information and be available to the client.

NOTE An audit conducted by use of remote auditing techniques is considered on-site.

### 9.1.4.2 Duration of the audit

Duration of the audit shall be a minimum of 80 % of the audit time. Time for audit reporting, audit planning or communication with the client shall not be more than 20 % of the audit time.

### 9.1.4.3 Audit days

The audit days are based on eight hours per day. Adjustments can be required based on local, regional or national legal requirements (e.g. inclusion of lunch breaks). The number of audit days shall not be reduced by using longer hours per working day.

NOTE The certification body can agree with the client the way in which the audit time is allocated. For example, if the audit time is four days:

- the audit can be performed during four consecutive or non-consecutive calendar days;
- the four calendar audit days can be spread over eight calendar half days;
- the audit team can be:
  - one auditor auditing during four calendar days; or
  - two auditors auditing individually during two calendar days each or any other similar combination.

If, after the calculation, the result is a decimal number, the number of days shall be adjusted either up or down to the nearest half day (e.g. 5,3 audit days becomes 5,5 audit days; 5,2 audit days becomes 5 audit days).

### 9.1.4.4 EnMS effective personnel

The number of EnMS effective personnel and the level of EnMS complexity, as defined in [Annex A](#), shall be used as the basis for the determination of audit time as shown in [Annex D](#). The certification body shall define and maintain documented information on a process for determining the number of EnMS

effective personnel for the scope of the certification and for each audit in the audit programme. The process for determining the number of EnMS effective personnel shall ensure that the persons who materially contribute to meeting the requirements of the EnMS are included.

### 9.1.5 Multi-site sampling

The requirements of ISO/IEC 17021-1:2015, 9.1.5, apply.

In addition, the certification of a multi-site organization based on sampling shall be allowed. The requirements of multi-site sampling as defined in [Annex B](#) shall be followed.

The certification body shall retain documented information on the decision criteria for sampling of the permanent sites and temporary sites. The criteria for determining the sample of the sites should be available to the client organization upon request.

### 9.1.6 Multiple management system standards

The requirements of ISO/IEC 17021-1:2015, 9.1.6, apply.

## 9.2 Planning audits

### 9.2.1 Determining audit objectives, scope and criteria

The requirements of ISO/IEC 17021-1:2015, 9.2.1, apply.

### 9.2.2 Audit team selection and assignments

The requirements of ISO/IEC 17021-1:2015, 9.2.2, apply.

### 9.2.3 Audit plan

The requirements of ISO/IEC 17021-1:2015, 9.2.3, apply.

## 9.3 Initial certification

### 9.3.1 Stage 1

The requirements of ISO/IEC 17021-1:2015, 9.3, apply.

Stage 1 shall include the following:

- a) a review of documented information on the scope and boundary(ies);
- b) a confirmation of the scope and boundary(ies) of the EnMS for certification;
- c) a confirmation of the number of EnMS effective personnel, energy types, SEUs and annual energy consumption, in order to review and confirm the audit time;
- d) a review of the documented information from the EnMS planning process;
- e) a review to confirm that EnPI(s) and corresponding EnB(s) are used by the client organization to determine energy performance;
- f) a review of documented information regarding determined and prioritized energy performance improvement opportunities as well as the objectives, energy targets and action plans.

Based on the outputs of Stage 1, the certification body shall confirm the competence needed for Stage 2.

### 9.3.2 Stage 2

**9.3.2.1** During Stage 2 audits, the audit team shall review the necessary audit evidence to determine whether or not continual energy performance improvement has been demonstrated prior to making a recommendation.

**9.3.2.2** The certification body shall analyse the necessary audit evidence to determine whether or not continual energy performance improvement has been demonstrated prior to making a certification decision. Confirmation of continual energy performance improvement shall be required for granting the initial certification.

NOTE Examples of how a client organization can demonstrate energy performance improvement are provided in ISO 50001:2018, A.10. Additional information on energy performance improvement is provided in [Annex C](#).

## 9.4 Conducting audits

### 9.4.1 General

The requirements of ISO/IEC 17021-1:2015, 9.4.1, apply.

In addition, energy performance improvement can be demonstrated at the equipment, process, system or facility level.

During each audit within the audit programme, the certification body shall confirm the suitability of the EnMS scope and boundary(ies) as defined by the client.

### 9.4.2 Conducting the opening meeting

The requirements of ISO/IEC 17021-1:2015, 9.4.2, apply.

### 9.4.3 Communication during the audit

The requirements of ISO/IEC 17021-1:2015, 9.4.3, apply.

### 9.4.4 Obtaining and verifying information

The requirements of ISO/IEC 17021-1:2015, 9.4.4, apply.

### 9.4.5 Identifying and recording audit findings

The requirements of ISO/IEC 17021-1:2015, 9.4.5, apply.

### 9.4.6 Preparing audit conclusions

The requirements of ISO/IEC 17021-1:2015, 9.4.6, apply.

### 9.4.7 Conducting the closing meeting

The requirements of ISO/IEC 17021-1:2015, 9.4.7, apply.

### 9.4.8 Audit report

The requirements of ISO/IEC 17021-1:2015, 9.4.8, apply.

In addition, an audit report shall include:

- a) the scope and boundary(ies) of the EnMS being audited;

b) at certification decisions:

- 1) a statement of the achievement of continual improvement of the EnMS with a record of audit evidence to support the statement;

NOTE 1 At the initial audit, the implementation of the system can be considered as continual improvement of the EnMS.

- 2) a statement of the achievement of continual energy performance improvement with a record of audit evidence to support the statement

NOTE 2 At the initial audit, the demonstration of energy performance improvement can be considered as continual energy performance improvement.

NOTE 3 See [Annex C](#) for additional information on energy performance improvement.

c) at surveillance audits, a statement confirming that the client organization has demonstrated implementation of actions for energy performance improvement (see [9.6.2](#)).

## **9.5 Certification decision**

### **9.5.1 General**

The requirements of ISO/IEC 17021-1:2015, 9.5.1, apply.

### **9.5.2 Actions prior to making a decision**

The requirements of ISO/IEC 17021-1:2015, 9.5.2, apply.

### **9.5.3 Information for granting initial certification**

The requirements of ISO/IEC 17021-1:2015, 9.5.3, apply.

In addition, the certification body shall review the necessary audit evidence to determine whether or not a continual energy performance improvement has been demonstrated prior to making an initial certification decision.

### **9.5.4 Information for granting recertification**

The requirements of ISO/IEC 17021-1:2015, 9.5.4, apply.

In addition, the certification body shall review the necessary audit evidence to determine whether or not a continual energy performance improvement has been demonstrated prior to making a recertification decision.

Confirmation of continual energy performance improvement shall be required for granting the recertification.

## **9.6 Maintaining certification**

### **9.6.1 General**

The requirements of ISO/IEC 17021-1:2015, 9.6.1, apply.

In addition, when conducting EnMS audits, the certification body shall ensure that, across the certification cycle, evidence related to the whole of the EnMS, including energy performance and energy performance improvement, is collected, evaluated and recorded as evidence in the audit reports.

NOTE See [9.3.2](#), [9.6.2](#) and [9.6.3](#) for evidence related to energy performance improvement.

### **9.6.2 Surveillance audit**

The requirements of ISO/IEC 17021-1:2015, 9.6.2, apply.

In addition, the client organization shall be able to demonstrate implementation of actions for energy performance improvement at the time of a surveillance audit. During the surveillance audits, demonstration of the achievement of energy performance improvement shall not be required (see [9.4.8](#)).

### **9.6.3 Recertification**

The requirements of ISO/IEC 17021-1:2015, 9.6.3, apply.

In addition, during recertification audits, the audit team shall review the necessary audit evidence to determine whether or not a continual energy performance improvement has been demonstrated prior to making a recommendation.

The recertification audits shall take into account any major change(s) including those in facilities, equipment, systems or processes.

NOTE Changes can result in the need to revise EnPIs or EnBs.

### **9.6.4 Special audits**

The requirements of ISO/IEC 17021-1:2015, 9.6.4, apply.

### **9.6.5 Suspending, withdrawing or reducing the scope of certification**

The requirements of ISO/IEC 17021-1:2015, 9.6.5, apply.

## **9.7 Appeals**

The requirements of ISO/IEC 17021-1:2015, 9.7, apply.

## **9.8 Complaints**

The requirements of ISO/IEC 17021-1:2015, 9.8, apply.

## **9.9 Client records**

The requirements of ISO/IEC 17021-1:2015, 9.9, apply.

## **10 Management system requirements for certification bodies**

The requirements of ISO/IEC 17021-1:2015, Clause 10, apply.

## **Annex A** **(normative)**

### **EnMS audit time**

#### **A.1 General**

This document provides specific requirements including those for the determination of audit time and sampling of multi-site organizations. The certification body shall implement this document when the requirements are different than those contained in other documents, except for scheme requirements or legal requirements.

In the case of accredited third-party certification, it is possible that the certification body will have to apply additional relevant documents (such as IAF Mandatory Documents) depending on the situation.

Audit time shall be determined based on a combination of the number of EnMS effective personnel described in [A.2](#) and a complexity factor, which is detailed in [A.4](#). The audit time can be adjusted due to a variety of factors that are discussed in [A.6](#).

#### **A.2 Determination of the EnMS effective personnel**

**A.2.1** When determining EnMS effective personnel, the process shall start with the entire set of potential persons including all permanent, full-time, temporary and part-time personnel.

Additionally, contractors or external service providers who either affect energy performance or affect energy performance improvement shall be considered.

Part-time personnel are based upon the hours worked. Part-time personnel numbers shall be converted to an equivalent number of full-time personnel (e.g. 30 part-time personnel working 4 h per day equates to 15 full-time personnel).

Thereafter, [A.2.2](#) is used to determine the EnMS effective personnel.

**A.2.2** The EnMS effective personnel shall be determined based on the process defined by the certification body. When the certification body defines their process for determining the number of EnMS effective personnel, the personnel who materially impact the energy performance and the effectiveness of the EnMS shall be taken into account, including the following:

- a) top management;
- b) the energy management team;
- c) the person(s) responsible for procurement related to energy performance;
- d) the person(s) responsible for making major changes that affect energy performance;
- e) the person(s) responsible for developing, implementing or maintaining energy performance improvements, including objectives, energy targets and action plans;
- f) the person(s) responsible for developing and maintaining energy data and analysis;
- g) the person(s) responsible for planning, operating and maintaining the processes related to the SEUs including during seasonal operations (e.g. harvesting activities, hotels) as appropriate;
- h) the person(s) responsible for design which affects energy performance.

NOTE It is important to understand the role and impact of personnel before excluding them as EnMS effective personnel.

**A.2.3** When reviewing the categories in [A.2.2](#), person(s) shall not be double counted (see [A.6](#)).

**A.2.4** Where a high percentage of EnMS effective personnel perform similar or repetitive processes, a reduction in the number is permitted. The justification and criteria for the determination of the EnMS effective personnel similar or repetitive processes shall be retained as documented information.

### A.3 Determination of energy types

The certification body shall require the client organization to provide the number of energy types that account for 80 % of the client organization's total energy consumption. The energy types are those identified in the energy review. This number is not necessarily the same as the number of energy types for the client organization.

For purposes of this document, energy types shall be those that cross the boundary(ies) of the EnMS.

NOTE 1 Energy types that are extracted (e.g. crude oil, gas, coal) or captured (e.g. solar, wind) within the client organization boundary(ies) are considered to enter the boundary at the point of extraction or capture.

NOTE 2 See ISO 50004 for additional detail on energy types.

### A.4 Determination of EnMS complexity

**A.4.1** The EnMS complexity is based on three criteria:

- annual energy consumption;
- number of energy types;
- number of SEUs.

**A.4.2** The EnMS complexity is a weighted and calculated value that addresses all three criteria listed in [A.4.1](#).

The complexity,  $C$ , is calculated using [Formula \(A.1\)](#):

$$C = (F_{EC} \times 0,25) + (F_{ET} \times 0,25) + (F_{SEU} \times 0,50) \quad (\text{A.1})$$

where

$F_{EC}$  is the annual energy consumption complexity factor from [Table A.1](#);

$F_{ET}$  is the number of the energy types complexity factor from [Table A.1](#);

$F_{SEU}$  is the number of the SEUs complexity factor from [Table A.1](#);

[Table A.1](#) provides the weighted value and the associated ranges for the factors needed to calculate the EnMS complexity.

**Table A.1 — EnMS complexity factors for determination of audit time**

| Criteria                                                                                                      | Weighted value | Range               | Complexity factor |
|---------------------------------------------------------------------------------------------------------------|----------------|---------------------|-------------------|
| Annual energy consumption (TJ)                                                                                | 25 %           | ≤ 20 TJ             | 1,0               |
|                                                                                                               |                | 20 TJ ≤ 200 TJ      | 1,2               |
|                                                                                                               |                | 200 TJ ≤ 2 000 TJ   | 1,4               |
|                                                                                                               |                | > 2 000 TJ          | 1,6               |
| Number of energy types                                                                                        | 25 %           | 1 to 2 energy types | 1,0               |
|                                                                                                               |                | 3 energy types      | 1,2               |
|                                                                                                               |                | ≥ 4 energy types    | 1,4               |
| Number of significant energy uses (SEUs)                                                                      | 50 %           | 1 to 3 SEUs         | 1,0               |
|                                                                                                               |                | 4 to 6 SEUs         | 1,2               |
|                                                                                                               |                | 7 to 10 SEUs        | 1,3               |
|                                                                                                               |                | 11 to 15 SEUs       | 1,4               |
|                                                                                                               |                | ≥ 16 SEUs           | 1,6               |
| NOTE The annual energy consumption and SEUs are those available from the client organization's energy review. |                |                     |                   |

**A.4.3** The complexity value *C* from [Formula \(A.1\)](#) is used to determine the level of EnMS complexity based on [Table A.2](#).

**Table A.2 — Level of the EnMS complexity**

| Complexity value <i>C</i> | Level of the EnMS complexity |
|---------------------------|------------------------------|
| > 1,35                    | High                         |
| 1,15 to 1,35              | Medium                       |
| < 1,15                    | Low                          |

## A.5 Determination of EnMS audit time

**A.5.1** The certification body shall determine the audit time based on a combination of the number of EnMS effective personnel and the level of EnMS complexity. The audit time for the initial certification (Stage 1 and Stage 2) is shown in [Table A.3](#). The certification body shall ensure that the audit time is reviewed and confirmed at Stage 1.

Where processes operate on a shift basis, the extent of auditing of each shift depends on the activities/ processes that take place on each shift and the level of control of each shift that is demonstrated by the client organization. To audit effective implementation, at least one of the shifts shall be audited. The method of sampling shifts and the justification for not auditing the other shifts shall be documented.

**Table A.3 — Initial certification audit time (audit days)**

| Number of EnMS effective personnel | Level of EnMS complexity                                                                                                                                                                                         |        |      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|
|                                    | Low                                                                                                                                                                                                              | Medium | High |
| 1 to 8                             | 2,5                                                                                                                                                                                                              | 4      | 5    |
| 9 to 15                            | 4                                                                                                                                                                                                                | 6      | 7    |
| 16 to 25                           | 5                                                                                                                                                                                                                | 7      | 9    |
| 26 to 65                           | 6,5                                                                                                                                                                                                              | 8      | 10   |
| 66 to 85                           | 8                                                                                                                                                                                                                | 9,5    | 11,5 |
| 86 to 175                          | 8,5                                                                                                                                                                                                              | 11     | 12   |
| 176 to 275                         | 9                                                                                                                                                                                                                | 11,5   | 12,5 |
| 276 to 425                         | 10                                                                                                                                                                                                               | 13     | 15   |
| ≥ 426                              | The certification body provides the audit time for a number of EnMS effective personnel exceeding 425. The certification body shall retain documented information on decisions made to calculate the audit time. |        |      |

**A.5.2** The audit time for the surveillance and recertification audits are shown in [Table A.4](#). The certification process shall ensure that major changes to the EnMS, SEUs, facilities, equipment, systems or processes results in a review of the determined audit time.

**Table A.4 — Surveillance and recertification audit time (audit days)**

| Number of EnMS effective personnel | Level of EnMS complexity                                                                                                                                                                                         |                 |              |                 |              |                 |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------|-----------------|--------------|-----------------|
|                                    | Low                                                                                                                                                                                                              |                 | Medium       |                 | High         |                 |
|                                    | Surveillance                                                                                                                                                                                                     | Recertification | Surveillance | Recertification | Surveillance | Recertification |
| 1 to 8                             | 1                                                                                                                                                                                                                | 1,5             | 1            | 2,5             | 1,5          | 3               |
| 9 to 15                            | 1                                                                                                                                                                                                                | 2,5             | 2            | 4               | 2,5          | 5               |
| 16 to 25                           | 2                                                                                                                                                                                                                | 3,5             | 2,5          | 5               | 3            | 6               |
| 26 to 65                           | 2,5                                                                                                                                                                                                              | 5               | 3            | 6               | 3,5          | 7               |
| 66 to 85                           | 2,5                                                                                                                                                                                                              | 6               | 3,5          | 6,5             | 3,5          | 8,5             |
| 86 to 175                          | 2,5                                                                                                                                                                                                              | 6               | 3,5          | 7               | 3,5          | 8,5             |
| 176 to 275                         | 3                                                                                                                                                                                                                | 6               | 4            | 8               | 4            | 9,5             |
| 276 to 425                         | 3,5                                                                                                                                                                                                              | 7               | 4            | 8,5             | 5            | 11              |
| ≥ 426                              | The certification body provides the audit time for a number of EnMS effective personnel exceeding 425. The certification body shall retain documented information on decisions made to calculate the audit time. |                 |              |                 |              |                 |

See [Annex D](#) for an example of minimum audit day calculation for an initial certification.

**A.5.3** Audits can include remote auditing techniques such as interactive web-based collaboration, web meetings, teleconferences and/or electronic verification of the client organization's processes.

**NOTE** Further information on the use of information and communication technology (ICT) for auditing/assessment purposes can be found in IAF MD 4:2018.

**A.5.4** Remote auditing activities shall be identified in the audit plan, and the time spent on these activities shall be considered as contributing to the duration of the audit. The audit plan shall include or reference the justification for the use of any remote auditing activities. It shall also include the selection of technologies and how they are managed.

**NOTE** Remote auditing can be used for other activities that are a part of audit time.

## A.6 Factors for adjustment of audit time

**A.6.1** The certification body shall provide the rationale for the decision and factors used to modify the audit time and ensure that it is retained as documented information. Factors for adjustment of the audit time can include:

- a) reduction:
  - 1) maturity of the management system;
  - 2) accredited verification claim related to energy performance improvement within the current certification cycle;
- b) increase:
  - 1) logistics and larger sites;
  - 2) multiple languages being used in the conducting of the audit;
  - 3) changes in the client organization;
  - 4) past audit findings;
  - 5) on-site generation of energy (e.g. generation of steam within the boundary, cogeneration);
  - 6) a person has multiple roles not accounted for in the EnMS effective personnel count.

**A.6.2** The reduction of audit time for an EnMS shall not exceed 30 % of the time established from [Tables A.3](#) and [A.4](#).

**NOTE** It is expected that duration of the audit is calculated at the end of the audit time determination (see [Annex D](#)).

**A.6.3** An audit of an integrated management system can result in increased audit time, but where it results in reduction it shall not exceed 20 %. The increase or decrease in audit time, for an integrated audit, is additional to any increase or decrease in audit time from [A.6.1](#) and [A.6.2](#) (see [Annex D](#)).

**NOTE** Further information on the application of ISO/IEC 17021-1 for audits of integrated management systems can be found in IAF MD 11:2013.

## A.7 Temporary sites

Where a client organization has temporary sites covering similar activities, the certification body shall use a multi-site sampling plan for the audit of its EnMS. Where the temporary sites do not cover similar activities, the temporary sites are not sampled (see [Annex B](#)).

## **Annex B**

### **(normative)**

## **Multi-site organizations**

### **B.1 General**

This document provides specific requirements including those for the determination of audit time and sampling of multi-site organizations. The certification body shall implement this document when the requirements are different than those contained in other documents, except for scheme requirements or legal requirements.

In the case of accredited third-party certification, it is possible that the certification body will have to apply additional relevant documents (such as IAF Mandatory Documents) depending on the situation.

This annex defines the requirements for the certification of an EnMS of client organizations with more than one site with a single EnMS. The approach defined shall ensure that the audits performed provide adequate confidence in the conformity of the EnMS across all sites listed and that the audits are practical, feasible and economical in operative terms. The EnMS shall demonstrate that it is capable of achieving the intended outcomes for all sites which are involved.

In this annex, references to the word “site” means permanent site (physical or virtual) or temporary site (physical or virtual), unless otherwise specified.

This annex does not cover multi-site organizations where multiple EnMS are deployed across the client organization, when each site shall be considered as an organization with a single site and audited accordingly.

Where a client organization's EnMS is subject to certification and the associated activities are carried out in a similar manner at different sites under the organization's authority and control, the certification body shall maintain documented information for sampling the sites at the initial certification audit, surveillance audit and recertification audits. Deviation from the requirements of [Annex B](#) can be considered if they are justified and evidence retained as documented information. Justification shall demonstrate that the same level of confidence in the conformity of the EnMS across all sites listed can be obtained prior to proceeding with the audits.

### **B.2 Sites in a multi-site organization**

#### **B.2.1 General**

A multi-site organization need not be a unique legal entity, but all sites shall have a contractual link with the central function of the client organization. The central function shall have the authority to require that the sites implement corrective actions when needed.

**NOTE** Where applicable, this can be set out in the formal agreement between the central function and the sites.

Where it is not practicable to define a site (e.g. for services), the coverage of the certification shall take into account the client organization's central function activities as well as delivery of its services. Where relevant, the certification body can decide that the certification audit shall be carried out where the client organization delivers its services and its central function shall be identified and audited.

### B.2.2 Temporary sites

Temporary sites may be included within the scope of the EnMS certification and included on the certification documents. Inclusion on the certification documents is subject to agreement between the client and the certification body.

## B.3 Eligibility of an organization for sampling

**B.3.1** The eligibility requirements for sampling are as follows.

- The client organization shall have a single EnMS.
- The client organization shall identify its central function. The central function is part of the client organization and shall not be subcontracted to an external organization.
- The central function shall have organizational authority to define, establish and maintain the single EnMS.
- The data appropriate for demonstrating energy performance is collected and can be analysed by the central function.
- The client organization's single EnMS shall be subject to a centralized management review.
- All sites shall be subject to the client organization's internal audit programme.

**B.3.2** The central function shall be responsible for ensuring that data (energy and other) from all sites is collected and analysed. It shall be able to demonstrate its authority and ability to initiate organizational change as required in regard to, but not limited to, the data given in [Tables B.1](#) and [B.2](#).

**Table B.1 — Management system data**

| Management system                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------|
| System documentation and system changes                                                                                |
| Management review                                                                                                      |
| Evaluation of corrective actions                                                                                       |
| Internal audit planning and evaluation of the results                                                                  |
| Demonstrate ability to collect information on legal requirements and other requirements and initiate changes as needed |

**Table B.2 — Energy performance data**

| Energy performance                                                                              |
|-------------------------------------------------------------------------------------------------|
| Consistent planning process                                                                     |
| Consistent criteria for determining, adjusting or revising EnB(s), relevant variables and EnPIs |
| Consistent criteria for establishing objectives, energy targets and action plans                |
| Centralized process for evaluating applicability and effectiveness of action plans and EnPIs    |
| Consistent criteria for evaluating energy performance improvement                               |

## B.4 Sampling methodology

### B.4.1 General

The sample shall be selected based on the criteria set out below. The certification body shall maintain documented information on the rationale for sample design and sample selection for sampling the sites at the initial certification audit, surveillance audit and recertification audits.

The certification body shall define the methodology for site selection and maintain documented information.

## **B.4.2 Conditions**

**B.4.2.1** An organization shall meet one or more of the following conditions to apply this annex:

- a) all sites are operated using similar activities or processes or SEUs;
- b) a certain number of sites can be organized into subsets that can be sampled where each site within the subset is operated using similar activities or processes or SEUs;
- c) several sites can be considered a single site if they are in close proximity to each other.

**EXAMPLE** A group of three sites in close proximity can be treated as a single site, in which case the number of EnMS effective personnel, types of energy, energy consumption and number of SEUs are combined.

If none of the criteria in this subclause apply, then all sites and the central function shall be audited.

**B.4.2.2** Where some of the sites under consideration have similar, but fewer, activities or processes than others, they can be eligible for inclusion under multi-site certification provided that the sites which perform the most energy intensive processes are subject to more frequent audits.

**B.4.2.3** The energy performance of the sites can be considered independently or as a whole. This shall be defined in the certification body's processes or in the justification for the multi-site organization's sampling plan (see [B.4.7](#)).

## **B.4.3 Site selection**

Site selection shall take into account the following criteria:

- a) results of internal site audits and management reviews or previous certification audits;
- b) significant variations in the size of the sites;
- c) variations in shift patterns and work process or procedures;
- d) complexity of the management system;
- e) processes conducted at different sites;
- f) modifications since the last certification audit;
- g) certification body's knowledge of the client organization;
- h) differences in language, and legal requirements and other requirements;
- i) geographical dispersion;
- j) complexity of energy types, energy consumption and SEUs;
- k) energy performance.

## **B.4.4 Selection of temporary sites**

The certification body shall retain information on the sites selected including any temporary sites which were operational and included in the sample.

The selection of temporary sites shall take into account:

- EnMS effective personnel;

- evaluation of the risks related to energy performance and energy performance improvement;
- energy consumption;
- types of energy that cross the boundary(ies) of the EnMS;
- variety of equipment, processes, systems or facilities and the various stages of projects;
- the transient nature of the sites.

If different criteria were applied to the temporary sites, a justification shall be retained as documented information.

### B.4.5 Size of the sample

The certification body shall maintain documented information for determining the sample size to be taken when auditing sites as a part of the audit and certification of a multi-site organization. This shall take into account the criteria described in this document. The certification body shall retain appropriate documented information for each application of multi-site sampling.

The minimum number of sites to be visited per audit shall be as follows.

- Initial certification audit: the size of the sample ( $Y$ ) shall be the square root of the number of sites ( $x$ ) rounded to the upper whole number, i.e.  $Y = \sqrt{x}$ .
- Surveillance audit: the size of the annual sample shall be the square root of the number of sites with zero point six as the coefficient, rounded to the upper whole number, i.e.  $Y = 0,6\sqrt{x}$ .
- Recertification audit: the size of the sample shall be the same as that for the initial audit.

Nevertheless, where the EnMS has proved to be effective over a period of three years, the sample size could be reduced by using a factor 0,8 rounded to the upper whole number, i.e.  $Y = 0,8\sqrt{x}$ .

### B.4.6 Risk

**B.4.6.1** The central function shall be audited during every initial certification and recertification audit and at least annually as a part of surveillance. The audit at the central function shall include a review of the energy performance from all the sites included in the entire organization certificate.

The sample size shall be increased or decreased where the certification body's risk analysis of the processes/activities covered by the EnMS subject to certification indicates special circumstances, such as the following:

- a) size of the sites and number of EnMS effective personnel;
- b) differences in working practices (e.g. shifts);
- c) differences in activities undertaken;
- d) differences in the energy consumption or SEUs;
- e) evidence of corrective action retained as documented information;
- f) applicable legal requirements or other requirements;
- g) results of internal audits and management review;
- h) the ability to demonstrate energy performance improvement and improvement of the EnMS.

**B.4.6.2** In an effort to reduce risk, the following conditions shall be met prior to the certification body's initial audit.

- a) The relevant sites (including the central function) shall be subject to the client organization's centrally managed internal audit programme prior to the certification body starting its audit process.

NOTE The internal audit can be conducted using remote electronic methods.

- b) The client organization shall have carried out a centralized management review of the EnMS prior to the certification body's initial audit.

## **B.4.7 Site sampling plan**

The certification body shall develop a site sampling plan for the audit programme.

The initial contract review process shall clarify which sites are to be included in the certification and which sites are to be excluded. If during the certification cycle sites are added or removed, the client organization shall notify the certification body. The sampling plan is adjusted as appropriate.

## **B.4.8 Additional sites**

When new sites join an already certified multi-site network, the new site or sites shall be considered as an independent set for the determination of the sample size. After inclusion of the new site or sites in the certificate, the new site shall be added to the existing ones for determining the sample size for future surveillance or recertification audits. If the number of sites is reduced, the sample size shall be reviewed to ensure that the sampling criteria can be demonstrated.

# **B.5 Audit and certification**

## **B.5.1 General**

The certification body shall maintain documented information to deal with audits under its multi-site programme. Documented information establishes the way the certification body confirms that the same EnMS governs the activities of all the sites, is applied to all the sites and all the eligibility criteria for the organization in [B.3](#) are met. The certification body shall justify and record the rationale for proceeding with any approach to the auditing and certification of a multi-site organization.

## **B.5.2 Application and application review**

The certification body shall obtain necessary information concerning the applicant organization to:

- confirm that a single management system is deployed across the client organization;
- determine the scope of the management system being operated and the requested scope of certification and, if applicable, sub-scopes;
- understand the legal and contractual arrangements for each site;
- understand “what happens where”, i.e. processes/activities/energy consumption/SEUs/energy performance improvement at each site, and identify the central function;
- determine the degree of centralization of processes/activities which are delivered to all sites (e.g. criteria for EnPIs);
- determine interfaces between the different sites;
- determine which sites are eligible for sampling and those that are not eligible (see [B.3](#));
- determine the audit time for the client organization;

- determine the competence required of the audit team(s).

### B.5.3 Audit programme

**B.5.3.1** In addition to the requirements in [9.1.3](#), the audit programme shall at least include or refer to the following:

- processes/activities/SEUs at each site;
- identification of those sites which are eligible to be sampled and which are not;
- identification of sites which are covered by sampling and which are not.

**B.5.3.2** When determining the audit programme, the certification body shall allow sufficient additional time for activities which are not part of the calculated audit time, such as travelling, communicating among audit team members, post-audit meetings, etc. due to the specific configuration of the client organization to be audited.

Remote auditing techniques may be used, provided that the processes to be audited are of such a nature that remote auditing is appropriate.

**B.5.3.3** Where audit teams consisting of more than one member are used at any point, it shall be the responsibility of the certification body, in conjunction with the team leader, to identify the technical competence, as defined in [Clause 7](#), for each part of the audit and for each site, and to allocate appropriate team members for each part of the audit.

**B.5.3.4** The audit time of the audit programme shall be the sum of the audit time at the central function and each site. In determining the audit time for the central function, the certification body shall take into account those EnMS requirements that are performed by the central function. The audit time for the central function and each selected site shall be calculated for each site using the audit timetables included in [Annex A](#). Evidence of the rationale for any adjustments (increase or decrease) and number of audit days shall be retained as documented information. For sites where actual processes and organizational structure are such that adjustments in audit time (increase or decrease) can be justified, the certification body shall provide the rationale for the decision and retain documented information.

### B.5.4 Calculation of audit time

**B.5.4.1** An organization that satisfies the eligibility criteria may consist of sites that can be sampled, sites that cannot be sampled or a combination of both. The audit time shall be sufficient to undertake an effective audit irrespective of the makeup of the client organization. The certification body shall justify any reductions applied and retain documented information as evidence.

**B.5.4.2** The audit time per selected site shall be calculated for each site, including any reductions followed by application of any applicable adjustments for considerations including integrated management systems, remote auditing, etc. The audit time for the organization shall be the sum of the audit time determined for each selected site and the central function.

### B.5.5 Audit plan

In addition to the requirement in [9.2.3](#), the certification body shall at least consider the following when preparing the audit plan:

- certification scope and sub-scopes for each site;
- energy types, SEUs, energy performance improvement to be audited;
- if multiple management system standards are being considered;

- processes/activities to be audited;
- audit time for each site;
- allocated audit team.

### **B.5.6 Initial audit**

During Stage 1, the audit team shall complete the information to:

- confirm the audit programme;
- plan Stage 2, taking into account the processes/activities to be audited at each site;
- confirm that the Stage 2 audit team has the required competence.

At the outcome of the initial audit, the audit team shall document which processes/activities were audited on each site visited. This information shall be used to amend the audit programme and audit plans for subsequent surveillance audits.

### **B.5.7 Nonconformities**

**B.5.7.1** When nonconformities, as defined in ISO/IEC 17021-1:2015, are found at any individual site, either through the organization's internal auditing or from auditing by the certification body, an investigation shall take place to determine whether the other sites can be affected. Therefore, the certification body shall require the client organization to review the nonconformities to determine whether or not they indicate an overall system deficiency applicable to other sites. If they are found to do so, corrective action shall be taken and verified both at the central function and at the individual affected sites. If they are found not to do so, the client organization shall be able to demonstrate to the certification body the justification for limiting its follow-up corrective action.

The certification body shall confirm that the client analyses the cause and describes the specific correction and corrective actions taken, or planned to be taken, to eliminate detected nonconformities, within a defined time including determining whether the other sites have been affected. The certification body shall require the client organization to review the nonconformities to determine if corrections or corrective action needs to be applied to the other sites. Evidence of the review and justification shall be retained as documented information.

**B.5.7.2** The certification body increases, as appropriate, its sampling frequency or the size of sample until it is satisfied that control has been re-established.

**B.5.7.3** At the time of the certification decision, if any site has a major nonconformity, certification shall be denied to the whole network of listed sites pending satisfactory corrective action.

**B.5.7.4** It shall not be admissible that in order to overcome the obstacle raised by the existence of a major nonconformity at a single site, the organization seeks to exclude the problematic site from the scope during the certification process.

### **B.5.8 Certification documents**

**B.5.8.1** Certification documents can be issued covering multiple sites provided that each site included in the scope of certification has either been individually audited by the certification body or audited using the sampling approach outlined in this document. The certification body shall provide certification documents to the client organization by any means it chooses.

**B.5.8.2** Certification documents can be issued to the client organization for each site covered by the certification under the condition that they contain the same scope, or a sub-scope of that scope, and include a clear reference to the main certification documents.

**B.5.8.3** The list of sites shall be kept updated by the certification body. To help ensure the accuracy of this information, the certification body shall request the client organization to inform it about the closure of any of the sites covered by the certification. Failure to provide such information shall be considered by the certification body as a misuse of the certification.

**B.5.8.4** Sites can be added to or removed from an existing certification. The certification body shall maintain documented information for the change in sites including revision of size of the sample, site selection and audit time calculations.

**B.5.8.5** The certification documents shall be withdrawn in their entirety if the central function or any of the sites do not fulfil the necessary provisions for the maintenance of the certification.

### **B.5.9 Surveillance audits**

Surveillance of multi-site organizations shall be based on the sampling methodology described in [B.4](#). The audit time per site shall be calculated in accordance with [Annex A](#). If the sampling eligibility criteria in [B.3](#) cannot be applied, the audit shall be based on the central function and 30 % of the sites. The sites selected for the second surveillance audit do not normally include the sites sampled as a part of the first surveillance audit.

### **B.5.10 Recertification audits**

Recertification of a multi-site organization shall be based on the sampling methodology described in [B.4](#). If the sampling eligibility criteria in [B.3](#) cannot be applied, then the central function and each site shall be audited. The audit time per site shall be calculated in accordance with [Annex A](#).

## **Annex C**

### **(informative)**

## **Energy performance improvement**

**C.1** When reviewing energy performance improvement for the purposes of making a recommendation at the certification decision point(s) in a certification cycle, the following issues are important to consider.

- Energy performance improvement shall be evaluated by comparing the EnPI value against the corresponding EnB (see ISO 50001:2018, 9.1.1). This can be done at a number of different levels including equipment, process, system or facility level.
- As indicated in ISO 50001:2018, A.4, demonstrating continual energy performance improvement across the scope and within the boundary(ies) of the EnMS does not mean all EnPI values improve. Some EnPI values improve and others do not; but across the scope of the EnMS, the organization demonstrates energy performance improvement.
- Similar to a single site, a multi-site organization can define energy performance improvement at several levels. That could include organization wide, by site, by system, by process or equipment. The data for energy performance improvement for the sampled sites as well as other sites is available through the central function.
- For a multi-site organization, not every site will contribute equally to the energy performance improvement of the multi-site organization. However, the data are expected to be available at the central function and confirmed at the sample sites.

**C.2** The organization can choose a number of methods including but not limited to ratios, linear regression, nonlinear regression, complex models, simulation, equipment and system test methods.

**C.3** The auditor should expect to see that for each EnPI the organization has done the following:

- a) collected energy consumption and potential relevant variable data at the level determined by the EnPI;
- b) determined which variables significantly affect energy performance and are, thus, relevant;
- c) established appropriate EnPIs using the relevant variables.

**C.4** If there are no relevant variables, then absolute energy consumption reduction can demonstrate energy performance improvement.

**NOTE** Additional information on energy performance metrics, EnBs and EnPIs can be found in ISO 50006 and ISO 50047.

## Annex D (informative)

### Examples of audit calculations

#### D.1 Example 1 — Audit time calculation

The number of EnMS effective personnel as determined by ABC certification body for XYZ company is 32.

The annual energy consumption reported is 12 TJ for a complexity factor of 1,0 and the weighted value is 25 % using [Table A.1](#).

The number of energy types (natural gas, electricity, diesel) is 3, using [Table A.1](#), the complexity factor is 1,2 and the weighted value is 25 %.

The number of SEUs for XYZ company is 3, using [Table A.1](#), the complexity factor is 1,0 and the weighted value is 50 %.

$$C = (0,25 \times 1,0) + (0,25 \times 1,2) + (0,50 \times 1,0)$$

$$C = 0,25 + 0,30 + 0,50$$

$$C = 1,05$$

Using [Table A.2](#), the complexity level is low since the complexity value is less than 1,15.

Using [Table A.3](#), the calculated audit time would be low complexity. Thirty-two EnMS effective personnel is 6,5 audit days for the Stage 1 and Stage 2 audit (see [9.1.4.2](#)).

Using [Table A.4](#), the calculated audit time for surveillance would be 2,5 audit days and for recertification would be 5 audit days (see [9.1.4.2](#)).

#### D.2 Example 2 — Single permanent site

a) Calculate the audit time from [Table A.3](#) for the initial audit.

1) Low complexity 16 EnMS effective personnel = 5 days.

b) Determine any reductions that apply and can be demonstrated.

1) The certification body applies its procedure PROC-01 revision 15, see [Table D.1](#).

NOTE [Table D.1](#) is an example excerpt from a fictional procedure and is not intended to represent requirements. The table is intended to provide context for the examples in [Annex D](#).

**Table D.1 — Reduction and addition adjustment factors**

| Item                                                                                                                         | Reduction | Addition |
|------------------------------------------------------------------------------------------------------------------------------|-----------|----------|
| Repetitive processes and organizational structure                                                                            | 10        |          |
| Maturity of the system                                                                                                       | 10        |          |
| Accredited ISO/IEC 17029 verification claim related to energy performance improvement within the current certification cycle | 20        |          |

Table D.1 (continued)

| Item                                                                                       | Reduction | Addition |
|--------------------------------------------------------------------------------------------|-----------|----------|
| Logistics and larger sites                                                                 |           | 5        |
| Multiple languages being used in the conducting of the audit                               |           |          |
| Person has multiple roles not accounted for in the EnMS effective personnel count          |           |          |
| Changes in the organization                                                                |           | 10       |
| Past audit findings                                                                        |           |          |
| On-site generation of energy (e.g. generation of steam within the boundary, co-generation) |           | 20       |

- 2) Repetitive process: A 3-D printing process is used and there are 98 identical machines: a 10 % reduction is applied.
  - 3) The site has on-site generation of energy: a 20 % addition is made.
  - 4) Total additions and reductions are +10 %.
  - 5) 10 % of 5 days is 0,5 days, the new total is 5,5 days.
  - 6) A check is made to see that the maximum deduction of 30 % is not exceeded: total deductions were 10 % which is less than 30 %.
- c) Is the EnMS a part of an integrated management system?
- 1) The EnMS system is being added to an existing integrated quality and environmental management system and it is integrated (see IAF MD 11:2013).
  - 2) Allowed reduction for integrated audit: based on Annex 2 level of integration and ability to perform integrated audit 10 %.
  - 3) Reduction allowed is 0,5 days, the new total is 5 days.
- d) Are there any temporary sites?
- 1) There are no temporary sites resulting in a change in audit time of zero (0).
- e) Are there any externally provided functions that need to be considered?
- 1) There are no externally provided functions that need to be considered, resulting in a change in audit time of zero (0).
- f) Calculate the duration of the audit from the determined audit time. This duration would be a minimum of 80 % of the determined audit time.
- 1) Four days on-site, 1 day off-site for audit preparation, communication and coordination and audit reporting activities.

### D.3 Example 3 — Multi-site organization

A hospitality organization requests a multi-site certification.

For purposes of this example, the questions shown in example 2 are not shown in detail. All the requirements of [Annex A](#) apply.

- a) Does the site qualify for multi-site sampling (i.e. do the various sites apply the same EnMS to all the sites being proposed for the multi-site certificate)?
  - 1) The review of this information indicates there is a central function which is the corporate office. The “corporate” EnMS applies to all of the sites. Each site is allowed to have additional documents as determined necessary to support the EnMS.
  - 2) The sites are all part of one corporation and have a clear relationship.
- b) How is energy performance addressed within the multi-site organization?
  - 1) The review of this information indicates that each site follows the same criteria and is responsible for energy performance improvement.
- c) How is the organization structured?
  - 1) The central function is performed at Site 1 which serves as both a call centre and a data centre.
  - 2) There are three permanent sites that are related to catering and food services. The activities of the three sites are the same, but the size of the sites varies. The locations are geographically located to cover North America, Europe and Australia.
  - 3) There are four permanent sites that are related to hotel activities. The activities and size of the four sites are the same at each location.
- d) Can a sampling plan be developed for this organization?
  - 1) Yes, based on this information the organization can be treated as three sub-sets and each sub-set sampled. The subsets and the data from all sites are shown in [Table D.2](#).

**Table D.2 — Subsets for sampling**

| Subset   | Site description                                              | Energy consumption<br>$T_j$ | Types of energy | SEUs | EnMS effective personnel |
|----------|---------------------------------------------------------------|-----------------------------|-----------------|------|--------------------------|
| Subset 1 | Call centre and data centre (central function performed here) | 1,5                         | 2               | 3    | 3                        |
| Subset 2 | North American site                                           | 0,4                         | 3               | 2    | 10                       |
|          | European site                                                 | 1,1                         | 2               | 3    | 20                       |
|          | Australian site                                               | 0,2                         | 2               | 1    | 5                        |
| Subset 3 | Hotel activities– 4 identical sites                           | 5                           | 3               | 2    | 7                        |

2) Sampling plan — Subset 2:

- i) In Subset 2, as per [Table D.2](#), the sample size is 3. Using [B.3.2](#), the number of sites to be visited for initial certification is calculated as 1,7 which is rounded off to 2. In Subset 2, the three sites show a wide variation in energy consumption and SEUs. As per [B.3.2](#), the sample size can be increased or decreased. In this case, a sample size of 2 is selected.
- ii) The choice of sites was made as per criteria in [B.3](#). The Australian site was selected by the certification body at random. The European site was selected by the certification body.

3) Sampling plan — Subset 3:

- i) In Subset 3, as per [Table D.2](#), the sample size is 4. Using [B.3.2](#), the number of sites to be visited for initial certification is calculated as 2. Since all sites are identical in respect of energy consumption and SEUs, the number of sites selected remains the same.

- ii) The choice of sites was made as per criteria in [B.3](#). The site “Sleep Well” was chosen at random. The site “Stay The Night” was selected by the certification body.
- e) For the subsets identified, are there some sites with fewer but similar processes that can be grouped together?
  - 1) Not applicable for this example.
- f) Determine the audit time for each site in the sample for the initial certification (Stage 1 + Stage 2). Detailed calculations are shown below for Subset 1 only.
  - 1) Subset 1:
    - i) Using [Formula \(A.1\)](#), complexity  $C$  is calculated as:
 
$$C = (1 \times 0,25) + (1 \times 0,25) + (1 \times 0,5) = 1,0$$
 which, for Subset 1 as per [Table A.2](#), the level of EnMS complexity is low.
    - ii) With three EnMS effective personnel and low complexity, the number of audit days as per [Table A.3](#) for initial certification is determined as 2,50.
    - iii) Based on PROC-01 revision 15, [Table D.1](#), of the certification body procedure, no adjustments are determined to be appropriate.
    - iv) The central function is located at this site so no adjustments are made.
    - v) The number of audit days is determined as 2,50 audit days.
  - 2) Subset 2 — European site:
    - i) Using [Formula \(A.1\)](#), complexity  $C$  is calculated as:
 
$$C = (1 \times 0,25) + (1 \times 0,25) + (1 \times 0,50) = 1,00$$
    - ii) Using [Annex A](#), for the European site as per [Table A.2](#), the level of EnMS complexity is low.
    - iii) With 20 EnMS effective personnel, which results in a value from [Table A.3](#) of 5 audit days and low complexity, the number of audit days as per [Table A.3](#) for initial certification are determined as 5,00.
    - iv) Based on PROC-01 revision 15, [Table D.1](#), of the certification body procedure, a 10 % reduction adjustment is determined to be appropriate for repetitive processes.
    - v) Additionally, the site does not need to be audited for the management system items audited at the central function. A 20 % reduction adjustment is considered to be appropriate.
    - vi) The overall reduction of audit time works out to 30 %. Based on this reduction, the audit time for the European site is determined as 3,50 audit days.
  - 3) Subset 2 — Australian site:
    - i) Using [Formula \(A.1\)](#), complexity  $C$  is calculated as:
 
$$C = (1 \times 0,25) + (1 \times 0,25) + (1 \times 0,50) = 1,00$$
    - ii) Using [Annex A](#), for the Australian site as per [Table A.2](#), the level of EnMS complexity is low.
    - iii) With five EnMS effective personnel and low complexity, the number of audit days as per [Table A.3](#) for initial certification are determined as 2,50.
    - iv) Based on PROC-01 revision 15, [Table D.1](#), of the certification body procedure, no adjustments are determined to be appropriate.

- v) Additionally, the site does not need to be audited for the management system items audited at the central function. A 20 % reduction adjustment is considered to be appropriate.
- vi) The overall reduction works out to 20 %. Based on this reduction, the audit time for initial certification of the Australian site is determined as 2,00 audit days.

4) Subset 3:

- i) Using [Formula \(A.1\)](#), complexity  $C$  is calculated as:

$$C = (1 \times 0,25) + (1,2 \times 0,25) + (1 \times 0,50) = 1,05$$

- ii) For the hotel sites, as per [Table A.2](#), the level of EnMS complexity is low.
- iii) With seven EnMS effective personnel and low complexity, the number of audit days as per [Table A.3](#) for initial certification are determined as 2,50.
- iv) Based on PROC-01 revision 15, [Table D.1](#), of the certification body procedure, a 10 % reduction adjustment is considered to be appropriate for repetitive processes.
- v) Additionally, the site does not need to be audited for the management system items audited at the central function. A 20 % reduction adjustment is considered to be appropriate.
- vi) The overall reduction works out to 30 %. Based on this reduction, the audit time for each site for initial certification is determined as 1,75 audit days.

g) Audit time for the multi-site organization.

- 1) The audit days for the multi-site organization are determined as shown in [Table D.3](#). The audit time for the multi-site organization works out to 11,50 audit days.

**Table D.3 — Determination of audit time**

| Subset   |                 | Number of sites | Audit time per site (audit days) | Audit time for subset (audit days) |
|----------|-----------------|-----------------|----------------------------------|------------------------------------|
| Subset 1 |                 | 1               | 2,50                             | 2,50                               |
| Subset 2 | European site   | 1               | 3,50                             | 5,50                               |
|          | Australian site | 1               | 2,00                             |                                    |
| Subset 3 |                 | 2               | 1,75                             | 3,50                               |
| Total    |                 | 5               |                                  | 11,50                              |

- 2) Are other reductions appropriate to this multi-site certificate?
  - i) Yes.
  - ii) IAF MD 11:2013 for integrated management systems. A reduction of 15 % based on the audit team and level of integration. This provides a reduction of 1,73 audit days.
  - iii) Total audit days = 9,77 audit days.
- 3) Are there any temporary sites or functions contracted externally?
  - i) Not applicable for this example.
- 4) The audit days are rounded off to 10. The minimum duration of the audit would be 8 audit days.

## Bibliography

- [1] ISO 9000:2015, *Quality management systems — Fundamentals and vocabulary*
- [2] ISO 19011:2018, *Guidelines for auditing management systems*
- [3] ISO/IEC/TS 17023:2013, *Conformity assessment — Guidelines for determining the duration of management system certification audits*
- [4] ISO 50002, *Energy audits — Requirements with guidance for use*
- [5] ISO 50004, *Energy management systems — Guidance for the implementation, maintenance and improvement of an ISO 50001 energy management system*
- [6] ISO 50006, *Energy management systems — Measuring energy performance using energy baselines (EnB) and energy performance indicators (EnPI) — General principles and guidance*
- [7] ISO 50015, *Energy management systems — Measurement and verification of energy performance of organizations — General principles and guidance*
- [8] ISO 50047, *Energy savings — Determination of energy savings in organizations*
- [9] ISO/IEC 17029, *Conformity assessment — General principles and requirements for validation and verification bodies*
- [10] IAF MD 4:2018<sup>1)</sup>, *IAF Mandatory Document for the Use of Information and Communication Technology (ICT) for Auditing/Assessment Purposes*
- [11] IAF MD 11:2013<sup>2)</sup>, *IAF Mandatory Document for Application of ISO/IEC 17021 for Audits of Integrated Management Systems (IMS)*

---

1) Available from: [https://www.iaf.nu/articles/Mandatory\\_Documents\\_/38](https://www.iaf.nu/articles/Mandatory_Documents_/38)

2) Available from: [https://www.iaf.nu/articles/Mandatory\\_Documents\\_/38](https://www.iaf.nu/articles/Mandatory_Documents_/38)

